(iv) A saving amount pays 8% interest per year, compounded continuously. In addition, the income from another investment is credited to the amount continuously at the rate of ₹ 400 per year.

2. Assume that a spherical rain drop evaporates at a rate proportional to its surface area. Form a differential equation involving the rate of change of the radius of the rain drop.

10.4.2 Formation of Differential Equations from Geometrical Problems

Given a family of functions parameterized by some constants, a differential equation can be formed by eliminating those constants of this family. For instance, the elimination of constants A and B from $y = \mathrm{A}\,e^{x} + \mathrm{B}\,e^{-x}$, yields a differential equation $\frac{d^2y}{dx^2} - y = 0$.

Consider an equation of a family of curves, which contains n arbitrary constants. To form a differential equation not containing any of these constants, let us proceed as follows:

Differentiate the given equation successively n times, getting n differential equations. Then eliminate n arbitrary constants from $(n+1)$ equations made up of the given equation and n newly obtained equations arising from n successive differentiations. The result of elimination gives the required differential equation which must contain a derivative of the nth order.

Example 10.2

Find the differential equation for the family of all straight lines passing through the origin.

Solution

The family of straight lines passing through the origin is $y = mx$, where m is an arbitrary constant. … (1)

Differentiating both sides with respect to x, we get

$$\frac{dy}{dx} = m. \qquad \ldots (2)$$

From (1) and (2), we get $y = x\frac{dy}{dx}$. This is the required differential equation.

Observe that the given equation $y = mx$ contains only one arbitrary constant and thus we get the differential equation of order one.

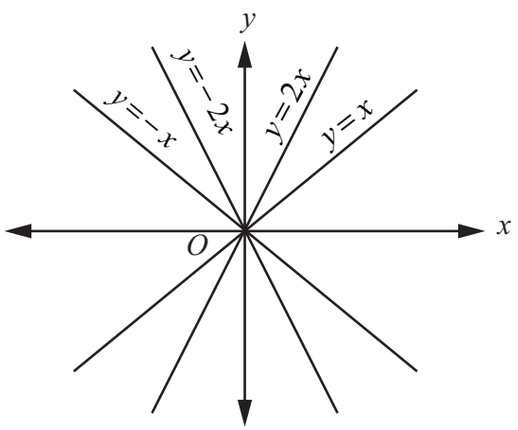


Fig. 10.1

Example 10.3

Form the differential equation by eliminating the arbitrary constants A and B from $y = \mathrm{A}\cos x + \mathrm{B}\sin x$.

Solution

Given that $y = \mathrm{A}\cos x + \mathrm{B}\sin x$ ... (1)

Differentiating (1) twice successively, we get

$$\frac{dy}{dx} = -\mathrm{A}\sin x + \mathrm{B}\cos x. \qquad \ldots (2)$$

$$\frac{d^2y}{dx^2} = -\mathrm{A}\cos x - \mathrm{B}\sin x = -(\mathrm{A}\cos x + \mathrm{B}\sin x). \qquad \ldots (3)$$

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